Managing Lufthansa's Dollar Exposure

A derivatives-based hedging analysis of the US$500 million Boeing order

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Context

In January 1985 Heinz Ruhnau committed Lufthansa to a US$500 million payment for twenty Boeing 737 aircraft, due on delivery a year later. The airline earned in deutsche marks, so the order created a clean one-year long-dollar liability: every pfennig the dollar gained against the mark widened the bill. At the January spot of 3.17 DM/USD the order was worth DM 1,585 million, and a move to 3.30 would have carried it to DM 1,650 million. The dollar had already run from 1.70 to 3.17 over five years and the interest differential pointed higher still, yet Ruhnau, with much of the G7 official community, read the currency as overvalued and due to correct. He expected 2.40 to 2.45 by delivery, could not time the turn, and feared 3.40 if it came late. Debt covenants limited his room to borrow. The problem was to hedge a large exposure under genuine two-sided uncertainty, not to trade a view he could not be sure of.

What follows prices each alternative, measures the risk it carries, and tests the criticisms later levelled at his choice. All computation sits in the accompanying workbook, driven from a single input sheet; figures are in DM millions unless stated, and each step carries its assumption where it is made. One convention holds throughout: the option premium is paid in January 1985, so it is future-valued to the January 1986 payment date at the DM deposit rate, which lets every strategy be compared on a single date.

1. Cost of each hedging strategy

The five structures behave as their construction implies. Staying uncovered is cheapest if the mark strengthens and dearest if it does not. The full forward strips out the movement and fixes the bill at DM 1,600 million wherever spot lands. The two partial forwards sit between those poles, each outlay a straight-line blend of the locked and the open portions. The full put hedge is the only structure that bends rather than tilts. It caps the cost at the strike when the dollar is strong and lets Lufthansa buy dollars in the market when the dollar is weak, so it floors the downside while keeping the upside Ruhnau expected. That asymmetry is what the premium buys.

Strategy	Spot 2.40	Spot 3.17	Spot 3.40
(a) Uncovered	1,200.0	1,585.0	1,700.0
(b) 100% forward	1,600.0	1,600.0	1,600.0
(c) 50% fwd / 50% open	1,400.0	1,592.5	1,650.0
(d) 75% fwd / 25% open	1,500.0	1,596.3	1,625.0
(e) Full put hedge	1,302.1	1,687.1	1,702.1

Final DM outlay, DM millions, at three realised spot rates. The full profile sits on sheet Q1 of the workbook.

Assumption. Partial hedges apply the 3.20 forward to the covered slice and realised spot to the remainder. The put pays 500 × min(spot, 3.20) plus the future-valued premium of DM 102.1m, and is exercised only when the dollar settles above the 3.20 strike.

2. Break-even between the option and the half-hedge

Setting the full put hedge equal to the 50 per cent forward and solving for spot gives a break-even of 2.79 DM/USD on a future-valued premium, or 2.82 if the premium is left undiscounted. Below that rate the option is the cheaper of the two; above it, up to and past the cap, the half-hedge wins. The figure matters because Ruhnau's own forecast of 2.40 to 2.45 sits well inside the option-favouring zone. On his stated view, the instrument he declined would have undercut the instrument he chose. That does not settle the decision, since he could not be certain of his forecast, but it sharpens the later charge that he reached for the wrong tool.

3. Risk and return across the strategies

Cost alone does not rank hedges; dispersion does. Measured across an equal-probability grid spanning his stated 2.40 to 3.40 band, the strategies separate cleanly. The open position carries the lowest expected cost and the highest risk. The full forward and the money-market hedge carry no risk at all, the latter at a materially lower locked cost. The partial forwards trade risk for cost close to linearly. The put is the instructive case: its cap pulls its standard deviation below the open position's, but the certain premium lifts its mean above both partial forwards. Insurance shows up in the metrics as a higher expected cost for a truncated tail.

Strategy	Mean	Std dev	Range
(a) Uncovered	1,450.0	176.8	500.0
(b) 100% forward	1,600.0	0.0	0.0
(c) 50% fwd / 50% open	1,525.0	88.4	250.0
(d) 75% fwd / 25% open	1,562.5	44.2	125.0
(e) Full put hedge	1,532.1	151.2	400.0
Money-market hedge	1,538.0	0.0	0.0

DM millions, on the equal-probability scenario grid. The 100% forward is dominated by the money-market hedge, which carries the same zero risk at a lower locked cost.

Assumption. Equal weights let the standard deviation read as pure dispersion rather than a restatement of Ruhnau's view. Weighting toward a weaker dollar would lower the mean of every open or participating structure without disturbing the risk ordering.

The efficient choices are the two ends: the money-market hedge for certainty at the lowest locked cost, and the open position for the lowest expected cost if the risk can be carried. The next section prices the first of those.

4. The money-market hedge

Covered interest parity turns the rate differential into a synthetic forward. Investing US$456.4m today at the 9.5625 per cent Eurodollar rate matures to exactly US$500m in a year. Buying that dollar amount now at the 3.17 spot costs DM 1,446.7m, funded by a DM loan that compounds at 6.3125 per cent to DM 1,538.0m at maturity. The effective rate locked is 3.076 DM/USD, which is precisely the parity forward, and it undercuts the bank's 3.20 forward by about DM 62m for identical certainty.

The reading is uncomfortable for the forward. At 3.20 the bank was quoting roughly twelve pfennigs rich to parity, so the certainty Ruhnau bought in the forward market was available more cheaply in the money

market. The caveat is real, and it is in the case: Lufthansa's covenants restricted new borrowing, which is exactly what this hedge requires. The cheapest certain route may therefore have been constrained or closed, and that is itself part of the defence of the route he did take.

Assumption. Lufthansa can lend and borrow at the quoted Euro rates and deal spot at 3.17, with no credit or transaction frictions.

5. Defending the half-hedge decision

With hindsight the mark strengthened, the hedged half cost more than leaving it open, and the 50 per cent forward duly lost. Judged from January 1985, on the information then to hand, the choice is more defensible than the outcome makes it look, for four reasons.

A view is not a certainty. Ruhnau believed the dollar would fall but admitted he could not time it and could see 3.40 if the turn came late. A half-hedge is the honest expression of a held but uncertain view: it wins something in every state and sidesteps the worst regret at both ends.

His duty was to the airline, not to his macro call. Leaving US$500m of a state-linked flag carrier open on a personal currency opinion is speculation, and it is hard to defend to a board whether or not it pays off. Hedging half separated the aircraft decision from a currency bet.

The covenant backdrop rewarded cost certainty on at least part of the exposure and constrained the cheaper money-market alternative, which narrowed the sensible menu to begin with.

The last point is about outcomes and evidence. The dollar's fall makes the open position look obvious now; it was not obvious then, when the rate differential and market momentum both pointed the other way. A decision is sound if the process was sound given what was known, and half-covering a large, uncertain, fiduciary exposure is a sound process.

6. Economics of the option premium

The option's appeal is its asymmetry; its difficulty is its price. At 6 per cent of the DM exercised, the premium is DM 96m upfront and DM 102m carried to delivery, about 6 per cent of the spot cost of the aircraft. That is a certain, visible cash cost booked in the current period whether or not the option is ever exercised.

The break-even against doing nothing is stark. The put only undercuts the open position once spot clears about 3.40, at or beyond Ruhnau's own worst case; in every milder outcome the premium buys protection that expires unused. Set against the forward, which costs nothing upfront, the option is really a paid-for right to participate if the dollar falls. A treasurer who has to justify a large certain outlay to a board, and book it now, often cannot, even when the flexibility is worth having.

Was it justified here? On Ruhnau's forecast, yes, and by a wide margin. At 2.40 the put costs about DM 1,302m against the forward's DM 1,600m, so a DM 102m premium buys roughly DM 300m of saving. The hesitation is not that the flexibility lacked value; it is that the value was contingent while the cost was certain, and boards reward certainty of cost.

7. An alternative option structure: the financed collar

The objection in the previous section is not to optionality but to its price, so the natural alternative keeps the protection and finances the premium. A collar, or cylinder, does precisely that. Lufthansa buys the 3.20 cap that protects against a rising dollar and, at the same time, sells a mark call, a dollar put, at a lower floor, using the premium it collects to pay for the cap. The effective purchase rate is then boxed into a band, protected above the cap and giving up participation below the floor.

Pricing needs a volatility, and rather than assume one I take it from the case itself. The volatility that makes a Garman-Kohlhagen call struck at 3.20 cost the quoted 6 per cent is about 20.9 per cent a year, which sits comfortably with how turbulent the mark-dollar rate had been. Using that single volatility to price both legs keeps the structure internally consistent with the premium the case already supplies.

Structure	Cap	Floor	Net premium upfront	Net premium at delivery
Collar A	3.20	2.90	15.7	16.7
Collar B	3.20	2.60	62.8	66.8
Plain put (Q1e)	3.20	n/a	96.0	102.1

Net premium in DM millions; legs priced by Garman-Kohlhagen at the backed-out 20.9 per cent volatility. The cap leg reprices the case option at DM 96m, which validates the volatility.

The effect is material. Selling a floor at 2.90 collects enough to cut the net premium from DM 96m to about DM 16m, roughly an eighty per cent reduction in the cost the treasurer has to defend. Dropping the floor to 2.60 keeps more of Ruhnau's expected upside and still nearly halves the premium.

The catch falls exactly where his conviction lay. A floor at 2.90 finances the cap cheaply but forfeits the 2.40 to 2.45 outcome he expected, the very scenario the option was meant to capture. So the collar's efficiency and his view pull against each other: the cheaper the structure, the more of his expected gain it signs away. The coherent version for a manager with his conviction sets the floor below his expected band, near 2.60, keeping most of the upside he believed in while still cutting the premium by half and capping the tail he feared. Against the Part 1 menu, that collar beats the plain put on cost for the same cap and beats the forward on flexibility, at the price of a floor he was, on his own forecast, unlikely to breach.

Assumption. European exercise, the backed-out 20.9 per cent volatility, continuous rates from the Euro deposit rates, and no transaction costs. Premiums scale with the US$500m notional.

8. Whether the board should retain Ruhnau

Four charges were laid. Together they read as a strong case; taken one at a time, against what was knowable in January 1985, most do not hold.

Start with the timing of the purchase at the dollar's peak. The peak was a peak only in hindsight. Aircraft are bought when the fleet needs them, and Lufthansa's did, with passenger traffic up seven per cent and freight up seventeen. The rate differential still favoured the dollar, and few called the top in real time. This is a hindsight judgment dressed as an analytical one.

The charge that he should have gone the whole hog, leaving the exposure open to match his conviction, fails on fiduciary grounds. A treasurer does not stake US$500m of a flag carrier on a personal forecast he admits he cannot time. Half-covering is the disciplined expression of an uncertain view, and the criticism inverts prudence into error.

The instrument criticism, forwards where options were available, carries more weight, though not for the reason usually given. Options would have preserved his expected upside, and a collar would have done so at a fraction of the premium. The sharper point the numbers expose is that the 3.20 forward was itself quoted rich to the 3.076 parity rate reachable through the money market, about DM 62m left on the table. Both better routes, the option and the money-market hedge, were blunted by the same covenant constraint, which stands as his mitigation.

The Boeing-versus-Airbus complaint belongs to industrial policy, not treasury. Lufthansa's stated policy was a mixed fleet, and Ruhnau was negotiating an Airbus order in parallel. Fleet commonality, delivery slots and commercial terms drive aircraft selection; the currency exposure that followed is a consequence of an operating decision, not a hedging error.

The recommendation is to keep him. The losses came from an unforecastable currency move, not a broken process, and a board that fires on outcomes rather than process teaches its managers to bury risk rather than manage it. His own defence is plain: he covered a fiduciary exposure he could not responsibly leave open, on a view he held honestly but could not time, under covenants that foreclosed the cheaper routes. The forward instrument, the numbers suggest, is the one element worth revisiting, and that is a lesson in execution rather than a cause for dismissal.

Recommendation

The decision was sound in process even though the outcome disappointed. The improvement worth carrying forward is instrument selection: options or a financed collar where flexibility has value, and the money-market route for certainty, each subject to the balance-sheet room the covenants allow.